Visual Search Experiment
PSY310 - Lab in Psychology
Lab Report

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GitHub Link:
https://github.com/aryanm4/PSY310_LabInPsychology/tree/Visual-Search

Introduction

Visual search tasks are one of the most basic methods used to effectively assess visual attention. Two parts of attention: selective and sustained attention can be identified in these tasks, as having a focus on specific stimuli placed among several distractors for a continuous period of time is required.[1] Efficiency in visual search tasks, i.e. the time it takes for one to accurately identify the target, is closely linked to one's attentive ability. Very poor accuracy rates and longer fixation times can also be an indicator of disorders related to attention, such as ADHD or learning disabilities.[2] Visual search insights are relevant to individuals' everyday behaviour because its applications are present in almost all daily tasks and laboratory research allows us to discover areas for refinement.[3]

[1] Labvanced.com
[2] Baghdadi et al, 2021
[3] Creem-Regehr, 2022

Method

Participants

The conducted experiment had 5 participants. They were all undergraduates students at Ahmedabad University between the ages of 18 and 22. There were 4 male participants and 1 female participant. They were fully informed of the experiment's objective, to find the target among the distractors as fast as possible, before it was conducted and full consent was provided.

Materials and Procedure

This experiment was created on a desktop computer with a resolution of 1920x1200 pixels using PsychoPy-2025.1.1. The procedure recorded the participants' reaction time taken in seconds to find the target among the distractors. 200 trials of the experiment were conducted per participant. When the trial began, the screen presented a cross-shaped fixation along with the target stimulus and several distractor stimuli. The target was a text in Open Sans font of the letter 'T' which was coded to randomly appear anywhere on the screen. The distractors was also a text in Open Sans font of the letter 'L.' The distractors were coded into 2 sets, in which the screen presented 5 or 10 distractors with a random probability anywhere on the screen, as shown in Figure 1. The participants were instructed to identify the 'T,' drag the pointer above it using the touchpad and click on it as fast as possible, after which the experiment would move onto the next trial. The responses were collected by the PsychoPy software on an Excel spreadsheet, which were then analysed to determine the average reaction time and accuracy.

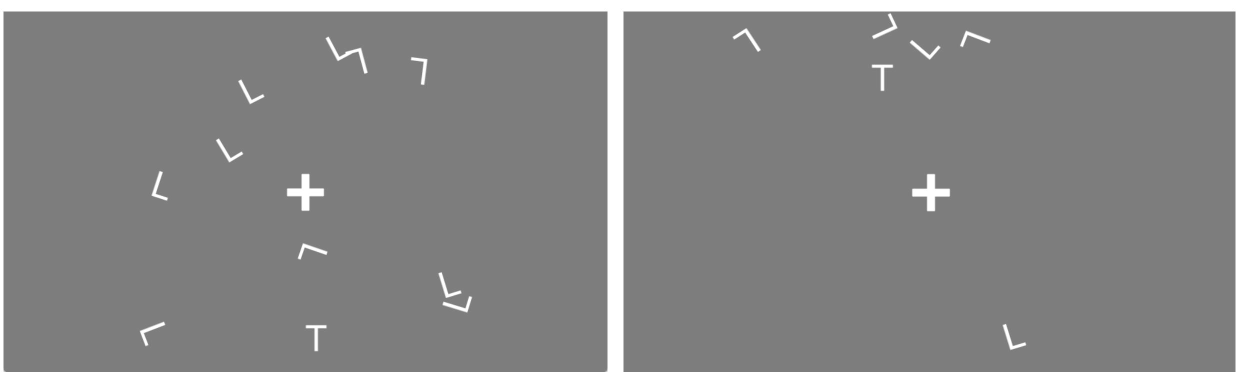

Figure 1 - Screen presenting target 'T' with 10 and 5 'L' distractors respectively.

Results

Across the 1000 trials performed by 5 participants, an average reaction time of 1.9685 seconds was obtained. A major insight found was that there is a significant difference between the reaction time when there are 5 distractors present versus when there are 10 distractors, for each of the participants. The slope was calculated by taking the difference of the average reaction times of both sets (2.2060 - 1.7166) and dividing it by the difference between the number of distractors in both sets (10 - 5). A value of 0.0979 seconds was calculated, which indicates that for every distractor that was added, a participant took an additional 0.0979 seconds to identify the target stimulus.

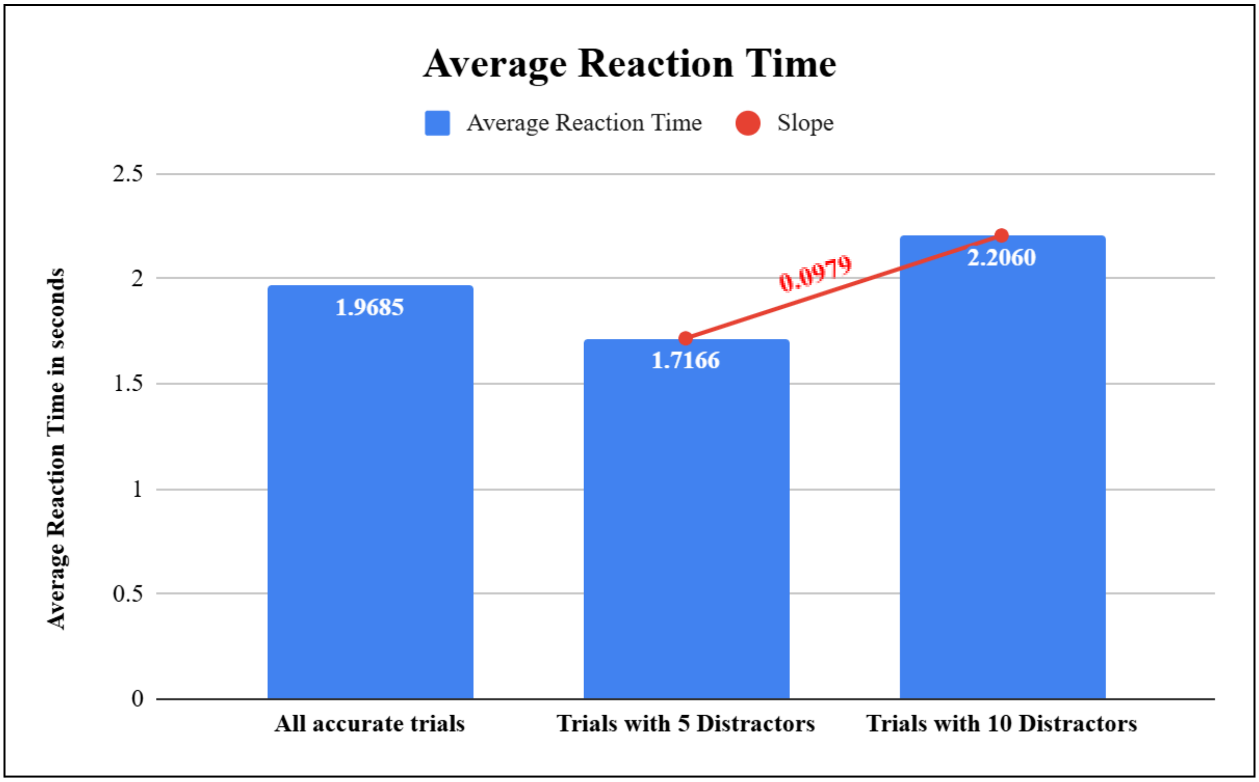

Figure 2 - Average Reaction Time of all participants

Discussion

After analysing the obtained data, the slope of reaction time by set size function was calculated to be 0.0979 seconds, which indicates that each distractor which was present on the screen delayed the reaction time to identify the target stimulus of a participant by 0.0979 seconds. This means that when a trial shifted from 5 to 10 distractors, participants' reaction time slowed down by approximately 0.49 seconds (0.0979 x 5). Therefore, a significant increase in reaction time is observed when the number of distractors is increased, which suggests that participants used a serial search method, in which they looked at each stimulus one-by-one to identify the target.

This visual search task has applicability in real-life situations to a certain degree. Identifying a stimulus among multiple distractors reflects several everyday tasks such as driving, shopping, etc. However, as this is done in a controlled environment with limited stimuli, such as only textual targets and distractors, we can not generalise the results for all real-life situations which have various sensory stimuli like noises and colours. Additionally, the sample size of 5 participants is not enough to be representative of any population. Therefore, further research is required.

References

Baghdadi, Golnaz, et al. “Chapter 7 - Assessment Methods.” Neurocognitive Mechanisms of Attention, Academic Press, 2021, pp. 203–50.

Creem-Regehr, Sarah. “Visual Search in Real-World and Applied Contexts.” Cognitive Research: Principles and Implications, 2022.

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